Transfer task: plan and begin

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Organize the given evidence and begin an independent position prediction.

SUCCESS CRITERIA - I can...

1. Organize the task data with fixed axes and matching units.
2. Calculate central and bounded final coordinates.
3. Save an explicit plan for the graph and explanation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “component” and “bound” to explain your model or result. Sentence starters are optional.

component: the part of a vector along one chosen axis

bound: an upper or lower limit in a model

assumption: a condition accepted for using a model

01 Bring back an earlier idea

A box at rest has 15 N right and 3 N left.
Find the additional horizontal force needed for balance and show the sum.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $x_f = x_i + v_x t$; $y_f = y_i + v_y t$.
Fictional fixed-Earth, constant-velocity model. CPA: x_i = -120, 000, 000 km; y_i = 0; t = 16, 000, 000 s; v_x = 8.21 km/s; v_y = 2.20 km/s.

Worked reasoning: A toy model uses x_i=0 m and constant v=2 m/s for 5 s, giving x=10 m. This is a conditional prediction, not proof that the real object follows the model.

Your turn: Draw the predicted path and mark one observation you would collect to test the constant-velocity assumption.

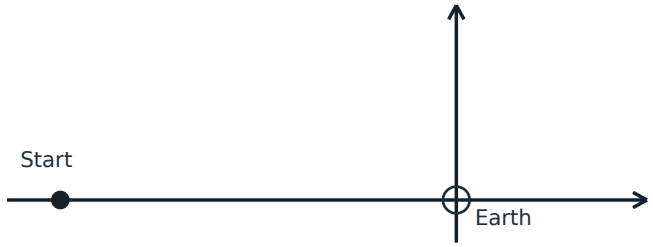
Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: ① ② ③

03 Organize the givens

Draw and label starting position, Earth at the origin, and both velocity directions. Create a table of given quantity, value and unit from the Reference paragraph.

Sentence starter: The ... in my model represents ... . Its label shows ... .



The small starting dot is the object; the labeled circle marks Earth at the origin. Not to scale.

Marzano check - circle: ① ② ③

04 Find the central end point

Calculate eastward coordinate change=8.21×16, 000, 000 km. Then add x_i.
Find the sideways coordinate change and final y.
Show one quantity per line.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

Transfer task: plan and begin

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Start the bounds

For CPA use provided independent bounds: x_i from -120, 050, 000 to -119, 950, 000 km; v_x from 8.02 to 8.40 km/s; v_y from 4.15 to 2.25 km/s. Time and y_i exact.
Calculate lower and upper final x and y.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

06 Choose a method

A rover travels straight east at a steady 7 m/s. A second rover turns at the same steady speed. Which can have zero net force?
Explain using velocity.

Word help: net force = the vector sum of all forces on one object

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

07 Checkpoint

Write a constant-velocity assumption and name one omitted physical effect. List what you finished and the specific next step for Day 22. Save all work.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Propose two different reasons a measured endpoint could be 8 m.
Give a measurement that would help distinguish them.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3